

CS 6120: Comparing Performance of Different Models on Empathy Detection

<https://github.com/sbolla2/NLP-Emotional-Classification>

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Abstract

As language model agents become more prominent in everyday life, model alignment becomes a more and more potent topic of research. How can we ensure that language models produce outputs that align with human values and expectations? One such method is to follow in the footsteps of many other issues in artificial intelligence, and have the models learn how to align themselves. In other words, can a language model learn to recognize empathy? Through this project, we present and compare three different approaches to learning to detect emotional intensity, emotion polarity, and empathy in human conversations. Of the models we've developed (an n-gram DAN, CNN, and BERT model), we found that the BERT model by far performs the best in capturing human empathy and emotion.

1 Introduction

Emotions, empathy, and distress are things that human beings experience without much effort. Simply as a reaction to a story or something that someone says, one might feel for the subject(s) of the story that they are hearing. Despite how easily this is done by people, the exact definitions and mechanics of empathy and emotion are nebulous and ill-defined. One understands them by feeling them for themselves. However, a machine does not feel emotions, so how does one codify these ideas that a human might express to a machine, trying to interpret them?

This question becomes increasingly important as more and more responsibility is given to language model agents. As time goes on, more and more people are turning to commercial LLMs for help with mental or physical health, therapy, substance recovery, etc. It is important to approach problems like these with empathy and tact, as one is advising someone in a sensitive position, and

a careless piece of advice could have disastrous consequences. Furthermore, tools like these are increasingly being used for information retrieval and as a substitute for traditional research. In situations like these, it is important for models to approach potentially sensitive or emotionally potent pieces of information in a way that aligns with human preferences, beliefs, and culture.

2 Background/Related Work

The issue of codifying emotion and empathy is the topic of the WASSA 2024 Shared Task on Empathy Detection and Emotion Classification and Personality Detection in Interactions ([was, 2024](#)). We focus on Track 2: Empathy and Emotion Prediction in Conversation Turns, which predicts empathy, emotion polarity, and emotion intensity at the turn level in conversations.

N-gram models provide a simple baseline for text classification by examining sequences of consecutive words. Iyyer et al. ([Iyyer et al., 2015](#)) proposed Deep Averaging Networks (DANs), which average word embeddings and pass them through a neural network. While computationally efficient, these models cannot capture long-range dependencies or complex semantic relationships, limiting their effectiveness for tasks requiring deeper understanding.

Convolutional neural networks (CNNs) have shown strong results for text classification. Kim ([Kim, 2014](#)) demonstrated that CNNs with pre-trained word embeddings perform well on sentiment analysis by using multiple filter sizes (3, 4, and 5 words) to capture different n-gram patterns. Shin et al. ([Shin et al., 2017](#)) applied CNNs with attention mechanisms to sentiment analysis on WASSA datasets. However, CNNs are limited to learning local patterns and miss the

broader context needed for understanding nuanced emotional content.

BERT (Devlin et al., 2019) uses bidirectional transformers and masked language modeling to learn contextual word representations from large text corpora. These word representations are pre-trained. This pre-training allows BERT to understand complex language patterns, which can then be fine-tuned for specific tasks with minimal changes to the model architecture. Liu et al. (Liu et al., 2019) improved on BERT with RoBERTa, which uses more training data and better training procedures. Li et al. (Li et al., 2021) showed that BERT effectively captures emotional information in text.

Previous work on empathy detection has demonstrated the effectiveness of transformer models. Dey and Girju (Dey and Girju, 2022) showed that BERT outperforms traditional machine learning methods for identifying empathy in medical narratives, achieving F1 scores above 0.70. For the WASSA 2024 shared task specifically, Yang et al. (Yang et al., 2024) achieved correlations of 0.544 for empathy and 0.644 for polarity using BERT with specialized training techniques, ranking 4th in the competition. These results show that well-designed transformer models can effectively predict empathy in conversational text.

3 Data

For our data, we are using the datasets provided by WASSA. This includes labeled test, train, and development sets. These datasets include 11167 training examples, 991 development examples, and 2317 test examples.

Each of the examples is an excerpt from a conversation between two people, where each person’s speaking turn is a unique example. The two people are conversing about an article or news story that the pair had recently read, in which there is harm to a person or group. Each of these turns of speech is annotated by a third party with Emotion, Emotional Polarity, and Empathy scores on a scale from 0 to 4 in $\frac{1}{3}$ increments. Each entry also has a Self Disclosure field where the speaker rates their own empathy on the same scale.

Table 1: (Abridged) Example Data Entry

text	Emotion	Polarity	Empathy	SelfDisc
"what..."	0.6667	1	0.6667	1

4 Methods

For this project, we will be comparing the results of three different models on each of three targets:

1. **Emotion:** How intense is the emotion the person speaking is feeling?
2. **Polarity:** How positive or negative is the emotion being felt.
3. **Empathy:** To what extent does the person speaking feel empathy towards the subject of conversation?

The first model we are evaluating is a deep averaging network using n-gram features. The second model we are evaluating is a CNN model. The final model we are evaluating is a BERT model (Devlin et al., 2019). A BERT model is the model most commonly used to approach this shared task. Entering this project, we hypothesized that BERT, with its pre-trained contextual understanding, would outperform the n-gram and CNN models. In particular, we expected the BERT model to perform well in empathy prediction, which requires deeper semantic understanding.

5 Experiments

The deep averaging network uses a set of learned n-gram features. To make a prediction, the model averages the n-gram embeddings of the input and passes them through a neural network.

In our CNN model, a set of learned filters are applied to learned word embeddings. This essentially generates unique n-gram embeddings. For the inference approach, these new embeddings are then passed through a neural network to predict a target.

Our BERT model is based on a BERT model that was pre-trained for regression tasks. We will be fine tuning the pre-trained model by adding a regression head on top of the BERT encoder. This will take the [CLS] token representation produced by the BERT model as input and output

a prediction for each of the three targets. During training, we are further fine-tuning all layers of the model. For inference, we pass the conversation turn through the BERT tokenizer and encoder, and use the final hidden state to generate predictions for emotional intensity, polarity, and empathy.

Each of the models will be trained on the provided training dataset using stochastic gradient descent with back propagation. The hyperparameters we support and utilize are referenced in Table 2.

Table 2: Tentative Hyperparameters for Model Training

Hyperparameter	N-gram/CNN	BERT
Learning Rate	1e-3	2e-5
Epochs	10	3
No. of Hidden Dimensions/Filters	100	768
Dropout	0.5	0.1
Batch Size	50	64
No. of N-Grams	3	N/A
Weight Decay	0.0001	0.01

To evaluate each of our models, we will be using the mean squared error and Pearson correlation coefficients of each model on each target.

6 Results

Table 3 presents the Pearson correlation coefficients, and Table 4 presents the mean squared error achieved by each model across all three prediction targets.

Target	Model	Test	Train	Dev
Empathy	N-gram	0.406	0.707	0.464
	CNN	0.510	0.891	0.594
	BERT	0.555	0.879	0.614
Polarity	N-gram	0.460	0.770	0.536
	CNN	0.509	0.909	0.602
	BERT	0.638	0.914	0.733
Intensity	N-gram	0.414	0.729	0.467
	CNN	0.538	0.900	0.583
	BERT	0.597	0.888	0.644

Table 3: Pearson correlation coefficients for all models

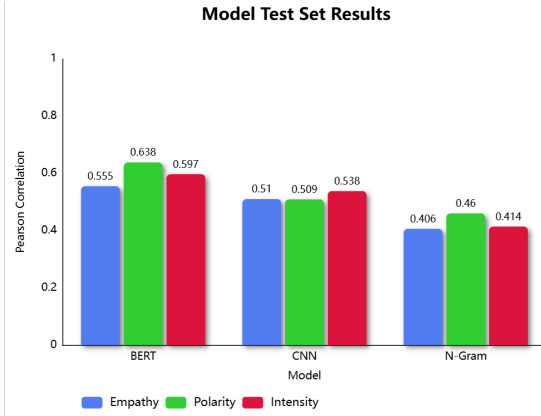


Figure 1: Pearson correlation on each target's test set grouped by model. From this, we can see that the BERT model outperforms the other two models for all three targets

Target	Model	Test	Train	Dev
Empathy	N-gram	1.338	0.539	1.070
	CNN	1.263	0.196	0.793
	BERT	1.289	0.192	0.766
Polarity	N-gram	0.374	0.153	0.348
	CNN	0.336	0.068	0.315
	BERT	0.264	0.064	0.227
Intensity	N-gram	0.814	0.327	0.515
	CNN	0.759	0.140	0.447
	BERT	0.718	0.148	0.383

Table 4: Mean squared error for all models

BERT significantly outperforms both baseline models across all three targets. On the test set, BERT achieved correlations of 0.555 (empathy), 0.638 (polarity), and 0.597 (intensity), compared to CNN's 0.510, 0.509, and 0.538, and n-gram's 0.406, 0.460, and 0.414. BERT also achieved the lowest mean squared errors for polarity (0.264) and intensity (0.718), while performing comparably to CNN for empathy (1.289 vs 1.263). The CNN model consistently outperformed the n-gram approach, demonstrating the value of learned convolutional features over simple averaging.

Performance varied across targets, with all models achieving highest correlations on polarity (BERT: 0.638), followed by intensity (BERT: 0.597), and empathy being most challenging (BERT: 0.555). This pattern suggests that polarity has clearer linguistic markers than the more nuanced concept of empathy. The MSE values support this finding, with polarity showing the

lowest errors across all models.

All models exhibited overfitting, with training correlations substantially higher than test correlations. For empathy, the train-test gaps were 0.301 (n-gram), 0.381 (CNN), and 0.324 (BERT). This overfitting reflects the challenge of training large models on limited data, with 11,167 training examples being relatively small for fine-tuning BERT’s 110 million parameters. Despite this overfitting, BERT’s test performance demonstrates that pre-trained representations provide reasonable generalization.

7 Conclusions

We compared three neural architectures for empathy detection on the WASSA 2024 shared task: n-gram averaging networks, CNNs, and BERT. BERT achieved the strongest performance across all targets, with test correlations of 0.555 (empathy), 0.638 (polarity), and 0.597 (intensity), outperforming CNN (0.510, 0.509, 0.538) and n-gram models (0.406, 0.460, 0.414). These results validate that pre-trained bidirectional transformers are superior for capturing empathetic language. Furthermore, our models demonstrate competitive performance compared to state-of-the-art WASSA 2024 systems, with an average Pearson correlation of 0.597, which slightly exceeds the 4th place system’s 0.595 Average Pearson correlation.

While our study establishes strong baselines, several promising directions for future research remain. Advanced transformer architectures such as RoBERTa (Liu et al., 2019) and DeBERTa offer potential performance improvements, as these models have demonstrated 2-5% improvement on similar computing tasks (Dey and Girju, 2022; Yang et al., 2024). Multi-task learning frameworks that jointly predict empathy, emotion polarity, and emotion intensity could exploit the relationships between these correlated variables to enhance overall performance. Addressing the overfitting observed in our experiments through data augmentation, ensemble methods, or advanced regularization techniques represents another important avenue for improvement. Finally, incorporating richer contextual features such as conversation history and speaker information, as well as exploring transfer learning from related empathy-annotated datasets, could further advance empathy detection

in conversational text. Pursuing these directions would build upon the strong foundation established by our BERT baseline and move toward more robust and accurate empathy detection systems.

Another interesting avenue to take this work would be in investigating societal biases captured in the models. Since the training data is explicitly focused on individuals’ reactions to stories about other people, one could engineer prompts to feed the model to investigate if the model shows more empathy or stronger emotions to one group over another. This could provide insight into emotional biases that exist in our society and effect how and when we choose to show compassion to others.

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